
Multi-Agent Orchestration for Electrocatalyst Discovery

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 The discovery of efficient hydrogen evolution reaction catalysts is hindered by
2 the combinatorial scale of candidate material space, which limits conventional
3 screening approaches. Here we present CatAgent, an autonomous multi-agent
4 workflow driven by large language models that achieves a 2.27-fold increase in
5 discovery rates over the random baseline. The framework integrates hypothesis
6 generation, surrogate-model evaluation, and iterative critic feedback to navigate
7 chemical space toward optimal hydrogen adsorption strength. We benchmark 13
8 state-of-the-art language models in 'single-shot mode' and 'iterative CatAgent
9 mode' across alloy compositions spanning $L1_0$ and $L1_2$ crystallographic phases.
10 Critic-enabled iterations improve performance for most model architectures, though
11 self-refinement effectiveness varies across model families. Top-performing models
12 such as Gemini 3 Flash and GPT-5.2 concentrate proposals near zero theoretical
13 overpotential, identifying successful candidates twice as efficiently as random
14 sampling. In contrast, outdated models such as GPT-3.5 Turbo fail to formulate
15 consistent hypotheses and exhibit unreliable search behavior. This work provides
16 the first demonstration that state-of-the-art language models can autonomously
17 reshape high-throughput catalyst screening into reasoning-guided exploration.

1 Introduction

18 Scalable renewable energy storage demands efficient electrocatalysts for hydrogen production, yet
19 state-of-the-art systems rely on scarce noble metals such as platinum and iridium [1]. This limita-
20 tion drives an intensive search for earth-abundant alternatives, with bimetallic alloys emerging as
21 promising candidates due to their tunable electronic properties. For the hydrogen evolution reaction
22 (HER), the hydrogen adsorption energy (ΔE_H) serves as a foundational descriptor for catalytic
23 activity, following a well-established volcano relationship where optimal performance correlates
24 with near-thermoneutral binding [2, 3]. Specifically, targeting a free energy of adsorption near zero
25 requires an electronic adsorption energy of approximately -0.27 eV, once zero-point energy and
26 entropy corrections are accounted for [4, 5].

28 However, navigating the vast chemical space of electrocatalysts presents a significant combinatorial
29 challenge [6]. Conventional high-throughput screening is computationally intensive, often requiring
30 exhaustive sampling of myriad adsorption sites, surface terminations, and adsorption configurations
31 [7]. Such enumeration typically relies on density functional theory (DFT), which remains prohibitively
32 expensive for large-scale exploration. To overcome these bottlenecks, researchers have increasingly
33 turned to active learning [8] and machine learning interatomic potentials [9], which can accelerate
34 discovery by an order of magnitude or more compared to manual exploration [10]. These frameworks
35 allow for rapid navigation through thousands of intermetallic configurations by prioritizing promising
36 candidates for high-fidelity evaluation.

Recently, the integration of large language models (LLMs) [11, 12] and transformer architectures [13] into materials science has introduced new paradigms for catalyst design [14–16]. Models such as CatBERTa and CatGPT have demonstrated that transformer-based representations can substitute expensive 3D descriptors in early-stage screening and enable inverse design of catalyst structures [17, 18]. Beyond these surrogate models, LLM-driven agentic workflows have emerged for fully autonomous experimentation. The A-Lab achieved a 63% success rate in synthesizing computationally predicted inorganic materials through closed-loop integration of ML-based synthesis planning, robotic experimentation, and active learning [19]. Similarly, multi-agent frameworks such as DREAMS have shown that LLM-based planners can orchestrate complex DFT simulation workflows with expert-level accuracy [20], while agentic systems for metal-organic framework discovery have demonstrated the capacity to autonomously navigate vast combinatorial spaces through iterative hypothesis generation and validation [21]. Frameworks such as Adsorb-Agent have further illustrated how LLM-driven reasoning can strategically identify stable adsorption configurations without exhaustive geometric enumeration by combining proposal engines with surrogate models such as graph neural networks [22]. These advancements demonstrate that LLM-based reasoning can drive autonomous navigation of chemical space, motivating the development of agentic workflows for catalyst discovery.

In this study, we investigate LLM-driven candidate selection for the screening of bimetallic HER catalysts within the $L1_0$ and $L1_2$ composition space. We compare the efficacy of a single-shot LLM approach against an iterative agentic workflow, CatAgent, with and without an integrated critic module for refined reasoning. Using a pretrained graph neural network potential (uma-s-1p1)[23] as surrogate model for rapid property evaluation, we assess how different LLM-orchestration strategies navigate bimetallic surface chemistry toward optimal hydrogen adsorption strength. The main contributions of this work are:

- A comprehensive benchmark of 13 state-of-the-art LLMs across three major providers (Google Gemini, OpenAI GPT, and Anthropic Claude), establishing quantitative metrics for chemical reasoning over 1,998 bimetallic compositions.
- CatAgent, a multi-agent framework integrating hypothesis generation, surrogate-model simulation, and critic feedback, achieving discovery rates up to 2.27-fold above random baseline.
- Systematic analysis revealing architecture-dependent search behaviors: top-performing models (Gemini 3 Flash, GPT-5.2) concentrate proposals near optimal adsorption energy, while outdated models (GPT-3.5 Turbo) exhibit dispersed exploration.

2 Methodology

2.1 Dataset

The search space for catalyst discovery is defined by bimetallic alloys composed of 37 metals: Al, Sc, Ti, V, Cr, Mn, Fe, Co, Ni, Cu, Zn, Ga, Y, Zr, Nb, Mo, Tc, Ru, Rh, Pd, Ag, Cd, In, Sn, La, Hf, Ta, W, Re, Os, Ir, Pt, Au, Hg, Tl, Pb, and Bi [24]. These metals are combined into two distinct composition families: the $L1_2$ structure with a stoichiometry of metal1:metal2 = 0.75:0.25, and the $L1_0$ structure with a stoichiometry of 0.50:0.50. The total candidate space consists of 1,998 unique entries, comprising 1,332 $L1_2$ and 666 $L1_0$ candidates. For each candidate, adsorption sites are enumerated according to symmetry, resulting in 9 unique sites for $L1_2$ and 10 for $L1_0$.

Adsorption energies are calculated for every enumerated site, and the representative value for a given composition is defined as the minimum adsorption energy across all sites. The adsorption energy ΔE_H is calculated relative to gas-phase H_2 as follows:

$$\Delta E_H = E_{\text{slab}+H} - E_{\text{slab}} - \frac{1}{2}E_{H_2}.$$

The target adsorption energy is -0.27 eV, corresponding to a near-thermoneutral free energy of adsorption ($\Delta G_H = \Delta E_H + 0.24$ eV ≈ 0) that maximizes HER activity according to Sabatier’s principle [4, 8]. A successful candidate is defined as falling within the window of -0.57 eV $\leq \Delta E_H \leq 0.03$ eV. The baseline dataset distribution corresponds to the minimum H adsorption energies across the 1,998 candidates, utilizing Universal Models for Atoms (UMA, uma-s-1p1 checkpoint) as a surrogate evaluator [23]. Among machine learning interatomic potentials benchmarked for hydrogen

87 adsorption energy prediction, UMA-s achieves the lowest total MAE (0.168 eV) combined with
 88 high structural reliability (84.58% normal relaxation rate), making it well-suited for rapid screening
 89 [25, 26].

90 2.2 CatAgent workflow

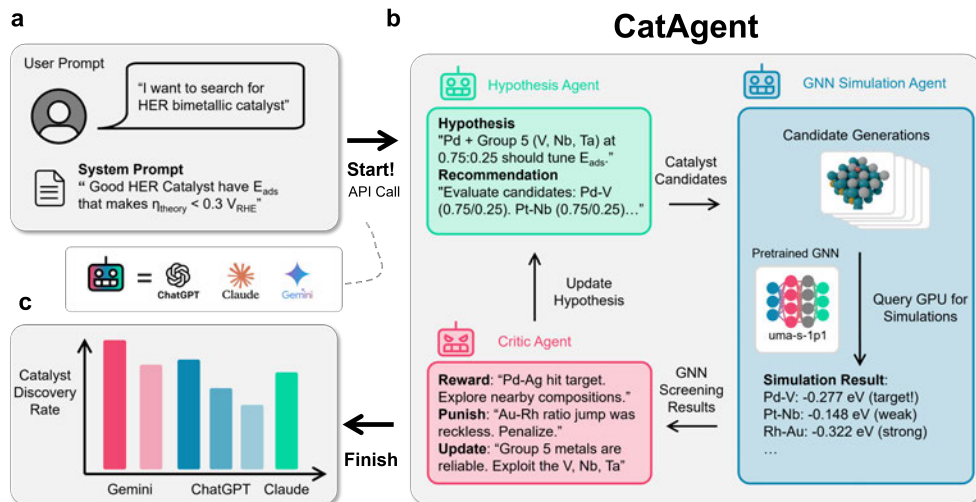


Figure 1: Schematic representation of the CatAgent multi-agent workflow architecture. (a) User and system prompts specify the screening objective and acceptance criterion. (b) Iterative loop of LLM-driven hypothesis formulation, candidate suggestion, UMA-based energy evaluation, and critic feedback for successive refinement. (c) Performance evaluation of screening outcomes.

91 Two primary screening approaches are employed: single-shot screening and the iterative CatAgent
 92 workflow. In the single-shot mode, the large language model (LLM) performs a combined role of
 93 hypothesis formulation and candidate generation to propose 200 candidates in a single non-iterative
 94 pass (Fig. 2a). This mode was evaluated using 13 models: Gemini 3 Flash, Gemini 3 Pro, Gemini 2.5
 95 Flash, Gemini 2.5 Pro, GPT-5.2, GPT-5, GPT-5 mini, GPT-5 nano, GPT-4o mini, GPT-3.5 Turbo,
 96 Claude Opus 4.5, Claude Haiku 4.5, and Claude Sonnet 4.5.

97 The CatAgent workflow utilizes a multi-agent architecture comprising LLM-driven hypothesis,
 98 simulation, and critic agents. After the user specifies the screening objective and acceptance criterion
 99 (Fig. 1a), each run explores 200 candidates over 20 steps, proposing 10 candidates per step. We
 100 investigate two variants of this workflow: a critic-enabled and a critic-disabled configuration. In
 101 the critic-enabled variant, the critic agent evaluates stepwise outcomes to identify recurring success
 102 and failure patterns, and its feedback is used by the hypothesis agent to refine the search strategy in
 103 subsequent steps (Fig. 1b). After termination of the iterative loop, overall screening performance
 104 is evaluated using the metrics defined in Section 2.3 (Fig. 1c), enabling systematic comparison of
 105 screening outcomes across models and workflow configurations. The workflow was benchmarked
 106 using Gemini 3 Flash, GPT-5.2, GPT-4o mini, and GPT-3.5 Turbo; for all models and providers,
 107 default sampling hyperparameters were used.

108 The simulation agent executes an evaluation pipeline using UMA: (i) construction of the bulk phase,
 109 (ii) bulk relaxation, (iii) construction of a (2×2) three-layer slab with 20 Å of vacuum, (iv) slab relax-
 110 ation, (v) adsorption site enumeration, (vi) relaxation of the H+slab adsorbed structure with the bottom
 111 two layers fixed, and (vii) calculation of ΔE_H using the gas-phase H_2 energy. Geometry optimizations
 112 were performed using the Limited-memory Broyden-Fletcher-Goldfarb-Shanno (LBFGS) algorithm
 113 with a force criterion of $f_{max} = 0.05$ eV/Å and a maximum of 100 steps. The codes, exact model
 114 snapshots, full prompts, and all raw outputs for the single-shot and CatAgent workflows are available
 115 at https://osf.io/t8he3/overview?view_only=1255c0a1cb1646bd9616d634314cd5e9.

2.3 Metrics and statistics

The performance of the screening process is evaluated using the success probability, defined as:

$$P_{\text{success}} = \frac{|\{i \mid \Delta E_{\text{H},i} \in [-0.57, 0.03] \text{ eV}\}|}{200}$$

where the success window corresponds to the target adsorption energy of $-0.27 \text{ eV} \pm 0.30 \text{ eV}$. The random-choice success probability is defined as the probability of selecting a successful candidate from the baseline distribution. Among the 1,998 candidates in the search space, 699 fall within the success window, yielding a random baseline of $699/1998 = 35.0\%$. The improvement factor is calculated as the ratio of the observed success probability to this random baseline; a value of 1.0 indicates performance equivalent to random selection, while values exceeding unity suggest an ability to identify compositions closer to the target adsorption energy.

Token usage is reported as the sum of input and output tokens. Cached tokens are excluded to ensure consistent comparison across providers where caching metadata may not be available. For $L1_0$ compositions, symmetric pairs are considered equivalent and deduplicated within and across steps. All statistical values derived from triplicate runs are reported as mean $\pm$ standard deviation.

3 Results and discussion

3.1 Intrinsic performance comparison of language models

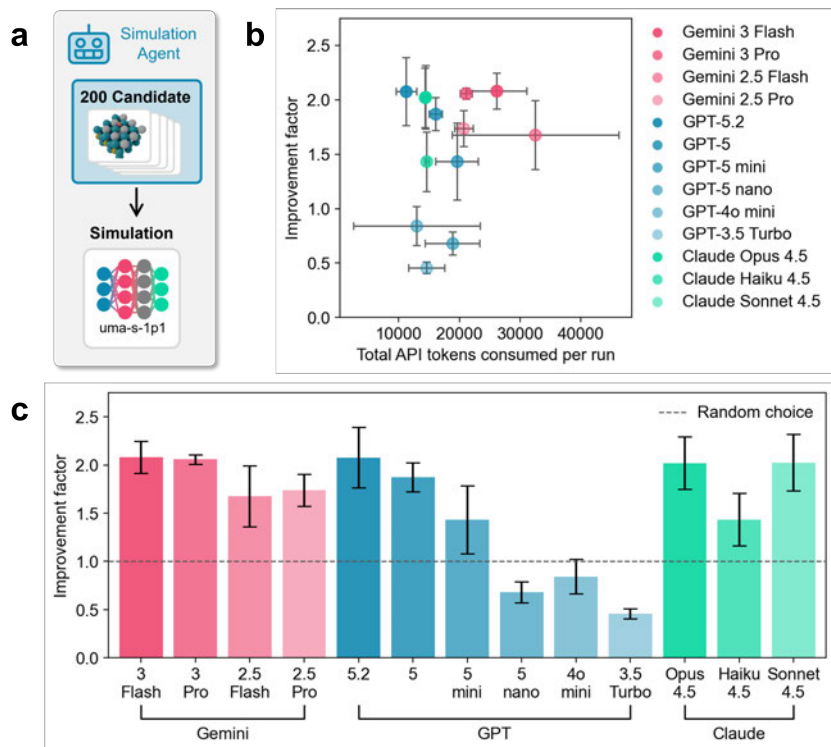


Figure 2: Single-shot screening performance analysis. (a) Workflow schematic. (b) Relationship between discovery efficiency and computational resource consumption across the 13 language models. (c) Improvement factors of the language models. Mean $\pm$ standard deviation from triplicate runs.

Single-shot benchmarking reveals substantial variation in chemical reasoning capabilities across the 13 language models evaluated (Fig. 2a). Resource consumption differs markedly among models, ranging from approximately 26,120 tokens per trial for Gemini 3 Flash to considerably lower values for more efficient architectures (Fig. 2b). Despite this variation in computational cost, performance differences are even more pronounced: Gemini 3 Flash achieves the highest mean improvement

factor of 2.08, demonstrating robust navigation of the chemical composition space in a single attempt, whereas GPT-3.5 Turbo exhibits an improvement factor of only 0.46, performing below the random baseline (Fig. 2c). This disparity across model architectures indicates that LLM training and design fundamentally determine single-shot predictive accuracy for catalyst discovery tasks.

Further analysis reveals that the number of unique candidates explored correlates strongly with the resulting improvement factor. While the experimental target is 200 unique catalyst candidates per trial, several models fail to meet this goal; GPT-4o mini averages only 116 candidates. Notably, resource efficiency does not track directly with overall performance: GPT-5.2 requires an average of only 78 tokens per successful candidate, representing a favorable balance between computational cost and discovery effectiveness. These metrics provide a comprehensive baseline for intrinsic model capabilities before introducing iterative agentic workflows.

3.2 CatAgent performance comparison across language models

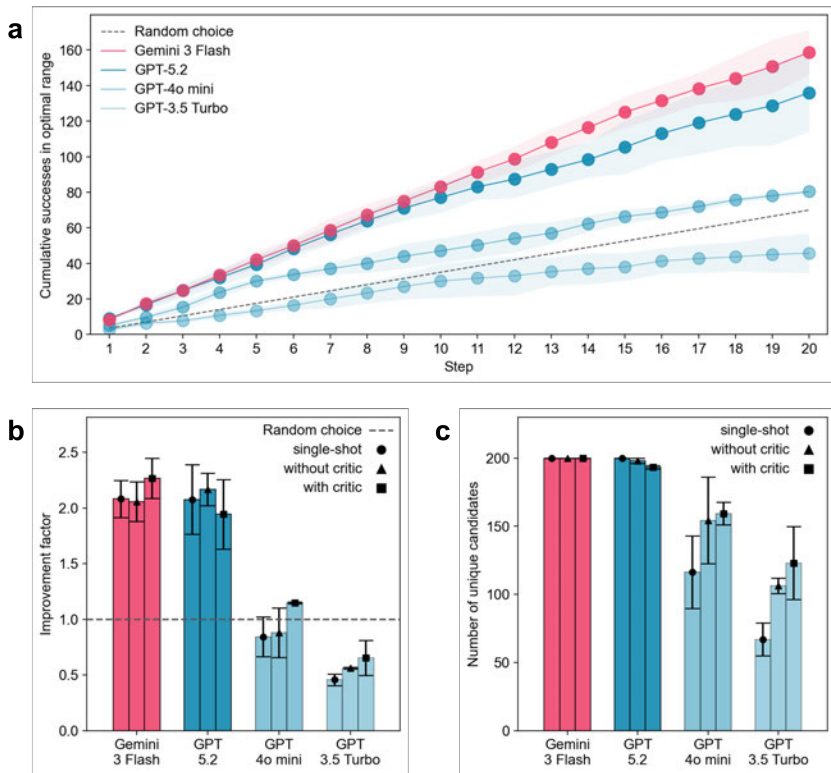


Figure 3: CatAgent workflow performance across model architectures. (a) Cumulative discovery of successful candidates during critic-guided iterative search. (b) Improvement factors and (c) unique candidate counts for single-shot, non-critic iterative, and critic-enabled iterative workflows across Gemini 3 Flash, GPT-5.2, GPT-4o mini, and GPT-3.5 Turbo. Mean $\pm$ standard deviation from triplicate runs.

The introduction of the CatAgent workflow allows for a direct comparison of discovery performance across different architectural designs: the single-shot baseline, the CatAgent workflow without critic, and the CatAgent workflow with critic. This analysis is focused on a subset of four models: Gemini 3 Flash, GPT-5.2, GPT-4o mini, and GPT-3.5 Turbo. As demonstrated in Fig. 3b, for Gemini 3 Flash, the addition of a critic module is consistent with an increase in the improvement factor of approximately 0.21 compared to the iterative workflow lacking a critic. Similarly, GPT-4o mini demonstrates a notable improvement in its improvement factor, rising from 0.88 to 1.15 when the critic module is integrated. These results suggest that iterative self-correction and the inclusion of an evaluation step generally enhance the model’s ability to narrow down the chemical search space toward the desired adsorption energy target.

158 However, the efficacy of the critic module is not universal across all model architectures. Contrary
 159 to the general trends observed in other models, GPT-5.2 exhibits a lower improvement factor when
 160 a critic is employed compared to iterative trials conducted without one. In some instances, success
 161 probability can fall below the level of a random choice, particularly for models like GPT-3.5 Turbo.
 162 This performance deficit can be attributed to unfocused search behaviors or a failure to generate
 163 the required number of unique candidates. Additionally, the integration of a critic can influence the
 164 volume of output (Fig. 3c); with a critic, GPT-4o mini explores approximately 159 unique candidates
 165 per trial, which represents an increase over its single-shot baseline of 116.

166 3.3 Analysis of catalyst compositions discovered via CatAgent

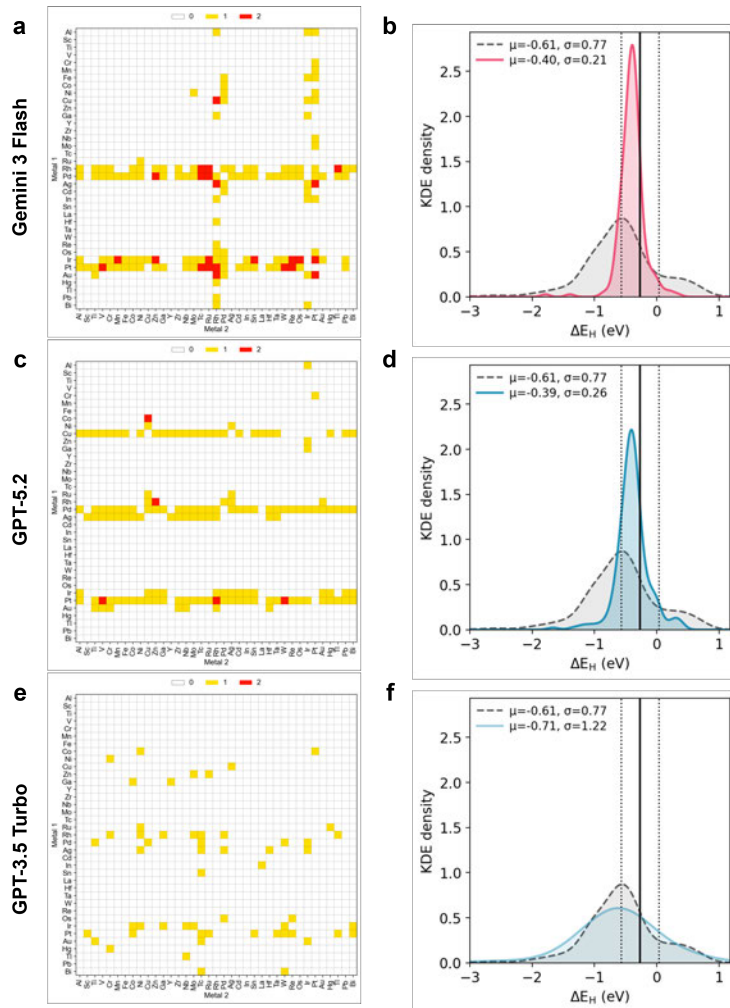


Figure 4: (a, c, e) Heatmaps showing the composition distributions of candidates with adsorption energies within the energy threshold and (b, d, f) KDE plots showing the adsorption-energy distributions of all catalysts identified in one repeat of the critic-enabled CatAgent. (a,b) Gemini 3 Flash; (c,d) GPT-5.2; (e,f) GPT-3.5 Turbo.

167 A qualitative analysis of the catalyst compositions discovered through the CatAgent workflows
 168 reveals different search behaviors, categorized as either focused or dispersed. The performance of the
 169 models is reflected in the adsorption-energy distribution mean and standard deviation relative to the
 170 broader dataset distribution. Models that exhibit focused search behaviors generate candidates with an
 171 energy distribution that has a smaller standard deviation and a mean shift toward the target, whereas
 172 dispersed search behaviors result in candidates that more closely resemble the original dataset's broad
 173 distribution.

174 Gemini 3 Flash exhibited a clear shift toward the target energy when using the CatAgent workflow
175 with critic (Figs. 4a,b, and Fig. 5). The mean absolute deviation from the target decreased from 0.34
176 eV at baseline to 0.12 eV across three trials. This focusing effect is also evident in the narrowing of
177 the energy distribution, with the standard deviation of adsorption energies reduced to an average of
178 36.7% of the baseline dataset value.

179 GPT-5.2 demonstrated the most precise alignment with the target energy among the models analyzed
180 (Figs. 4c,d, and Fig. 6). In the CatAgent workflow with critic, GPT-5.2 achieved a mean absolute
181 deviation of 0.08 eV from the target. The model shifted the mean candidate adsorption energy by
182 0.26 eV relative to the dataset baseline and reduced the standard deviation to 47.4% of the dataset
183 value. This combination of a significant mean shift and a reduced standard deviation indicates that
184 GPT-5.2 employs a focused search strategy that identifies compositions near the desired adsorption
185 energy.

186 In contrast, GPT-3.5 Turbo exhibited dispersed search behavior with limited success in approaching
187 the HER target (Figs. 4e,f, and Fig. 8). In the CatAgent workflow with critic, GPT-3.5 Turbo shifted
188 the mean candidate adsorption energy by -0.05 eV relative to the dataset baseline across three trials,
189 representing a move away from the target direction. Coupled with its reduced candidate generation
190 count of 123, the resulting energy distribution remained largely unfocused, consistent with its lower
191 normalized success rate. The contrasting energy distribution characteristics across these models
192 highlight the varying capacity of language models to interpret and act upon chemical design targets
193 within an agentic framework.

194 4 Conclusion

195 CatAgent, a multi-agent workflow driven by large language models, demonstrates substantial effective-
196 ness in navigating complex chemical spaces for catalyst discovery. Through systematic evaluation
197 of 13 language models in single-shot screening and four models within iterative multi-agent configura-
198 tions, we establish that LLM-guided exploration can strategically identify promising bimetallic
199 hydrogen evolution reaction catalysts from a search space of 1,998 compositions spanning $L1_0$ and
200 $L1_2$ crystallographic phases.

201 The CatAgent framework integrates hypothesis generation, computational evaluation, and critic
202 feedback mechanisms to achieve discovery rates up to 2.27-fold above random baseline selection.
203 The integration of critic modules demonstrates architecture-dependent efficacy, with models such as
204 GPT-4o mini exhibiting substantial performance improvements while GPT-5.2 showing decreased
205 effectiveness, highlighting the heterogeneous nature of self-refinement capabilities across model fam-
206 ilies. Notably, performance remains robust across three major API providers—Google, OpenAI, and
207 Anthropic—validating the generalizability of the approach beyond specific model implementations.
208 Leading models like Gemini 3 Flash demonstrate focused search behaviors that concentrate candidate
209 proposals near the target adsorption energy while substantially reducing distribution variance relative
210 to the baseline dataset. Resource efficiency varies significantly across LLM architectures, with
211 optimal configurations requiring as few as 78 tokens per successful candidate.

212 However, several opportunities for advancement remain. First, the current search space is relatively
213 constrained. Scaling to broader compositional ranges, alternative crystal structures, and multi-
214 component systems would better test framework capabilities. Second, whether domain-specific
215 fine-tuning on materials science data could enhance chemical reasoning remains unexplored. Third,
216 extending CatAgent to multi-objective optimization would enhance practical utility, as real-world cat-
217 alyst design requires balancing activity, selectivity, stability, cost, and synthesizability simultaneously.
218 This work establishes LLM-powered autonomous reasoning as a promising paradigm for accelerating
219 electrocatalyst discovery in vast chemical spaces. Continued development in LLM reasoning capacity,
220 prompting strategies, and experimental integration will be essential for realizing the full potential of
221 AI-assisted materials discovery.

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A Appendix

A.1 List of Assets

- **LangChain** [27] [MIT License]
- **LangGraph** [28] [MIT License]
- **FAIR-Chem** / fairchem-core [29] [MIT License]
- **UMA pretrained potential** (uma-s-1p1) [30] [FAIR Chemistry License v1]
- **OpenAI API + GPT-family model endpoints** [31] [Proprietary (API Terms of Use)]
- **Google Gemini API endpoints** [32] [Proprietary (API Terms of Service)]
- **Anthropic Claude API endpoints** [33] [Proprietary (API Terms of Service)]

A.2 Compositional distribution and energetic profiles of catalysts identified via CatAgent with critic

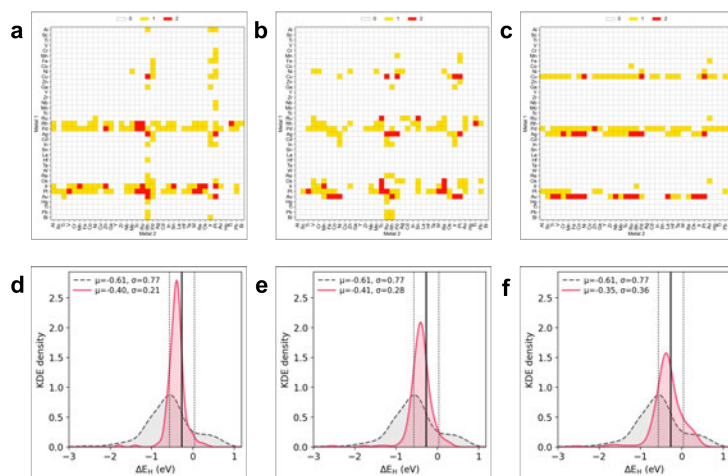


Figure 5: Catalyst discovery patterns for Gemini 3 Flash using critic-enabled CatAgent across three independent trials. (a-c) composition heatmaps of successful candidates; (d-f) kernel density estimates of adsorption energy distributions for all explored candidates. (a,d) Trial 1, (b,e) Trial 2, (c,f) Trial 3.

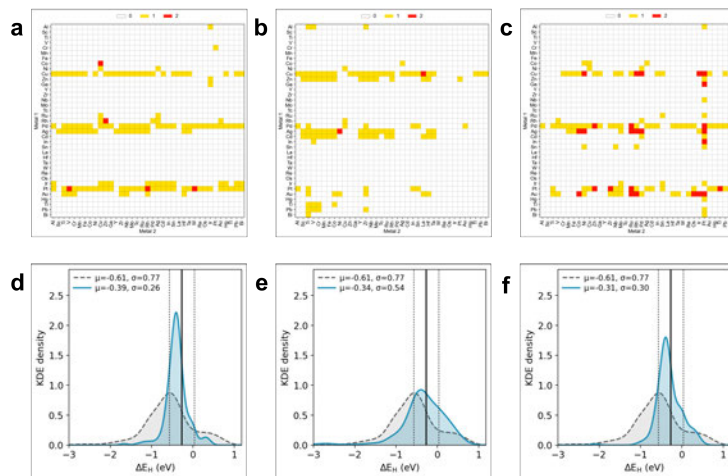


Figure 6: Catalyst discovery patterns for GPT-5.2 using critic-enabled CatAgent across three independent trials. (a-c) composition heatmaps of successful candidates; (d-f) kernel density estimates of adsorption energy distributions for all explored candidates. (a,d) Trial 1, (b,e) Trial 2, (c,f) Trial 3.

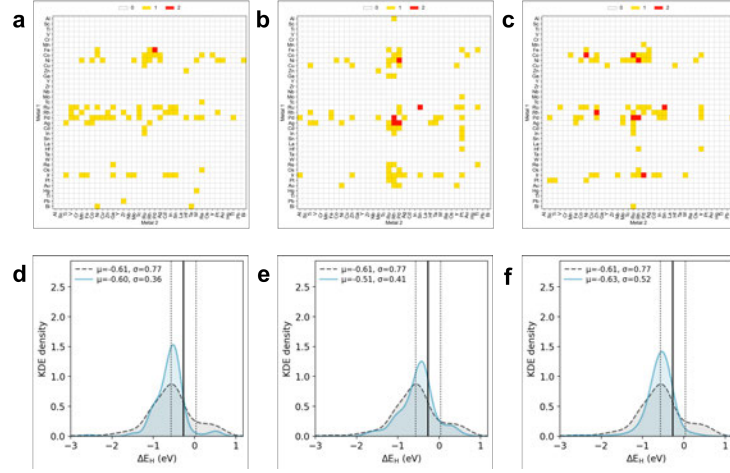


Figure 7: Catalyst discovery patterns for GPT-4o mini using critic-enabled CatAgent across three independent trials. (a-c) composition heatmaps of successful candidates; (d-f) kernel density estimates of adsorption energy distributions for all explored candidates. (a,d) Trial 1, (b,e) Trial 2, (c,f) Trial 3.

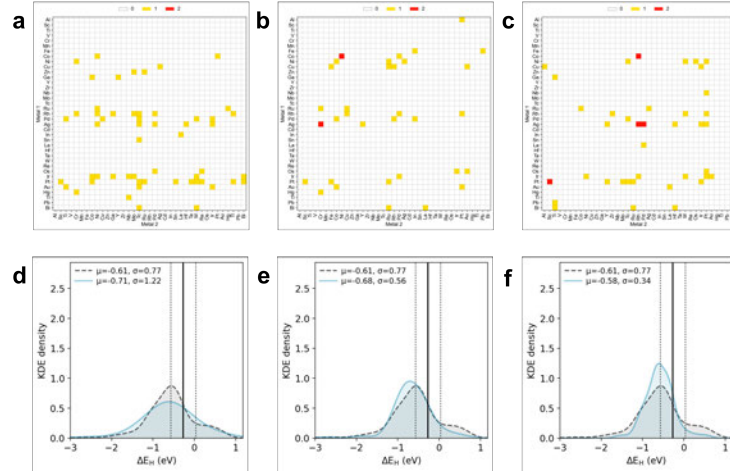


Figure 8: Catalyst discovery patterns for GPT-3.5 Turbo using critic-enabled CatAgent across three independent trials. (a-c) composition heatmaps of successful candidates; (d-f) kernel density estimates of adsorption energy distributions for all explored candidates. (a,d) Trial 1, (b,e) Trial 2, (c,f) Trial 3.

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